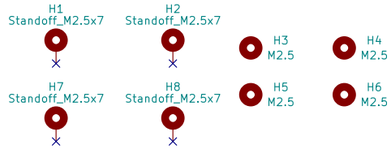
LINK-BASE

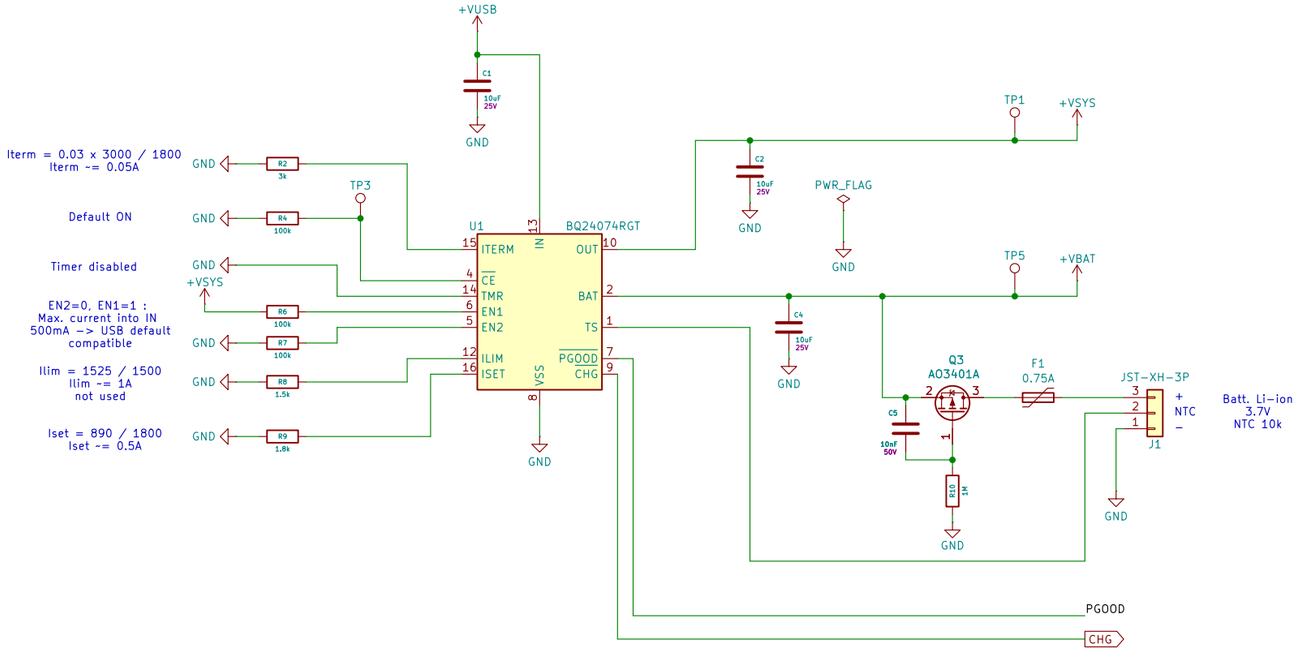


Sheet: /link-base-brd/
File: link-base-brd.kicad_sch

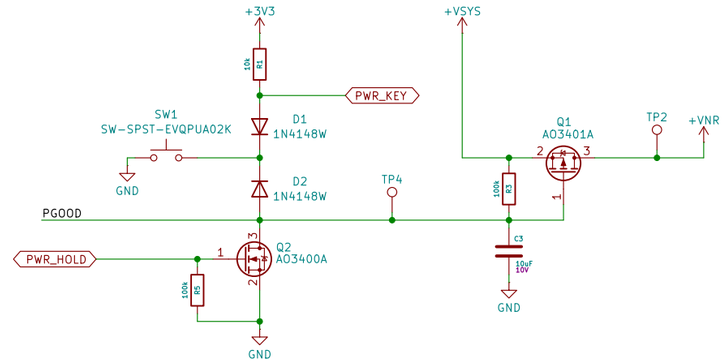
Title: Link Base Board

Size: A3	Date: 2026-08-28	**Rev: 0.2.0**
KiCad E.D.A. 10.0.5		Id: 1/6

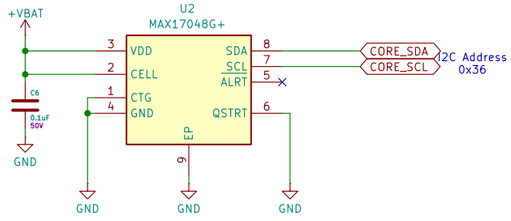
LI-ION CHARGER



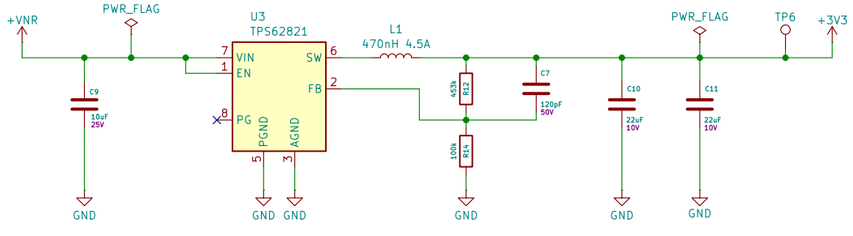
PWR LATCH



BATTERY JAUGE

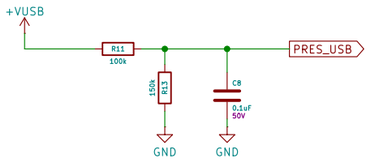


DC/DC 3V3@1A



USB DETECTION

5V -> 3V



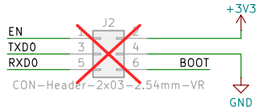
PWR

Sheet: /PWR/
File: pwr_supply.kicad_sch

Title: Link Base Board

Size: A3	Date: 2026-08-28	**Rev: 0.2.0**
KiCad E.D.A. 10.0.5		Id: 2/6

FLASH



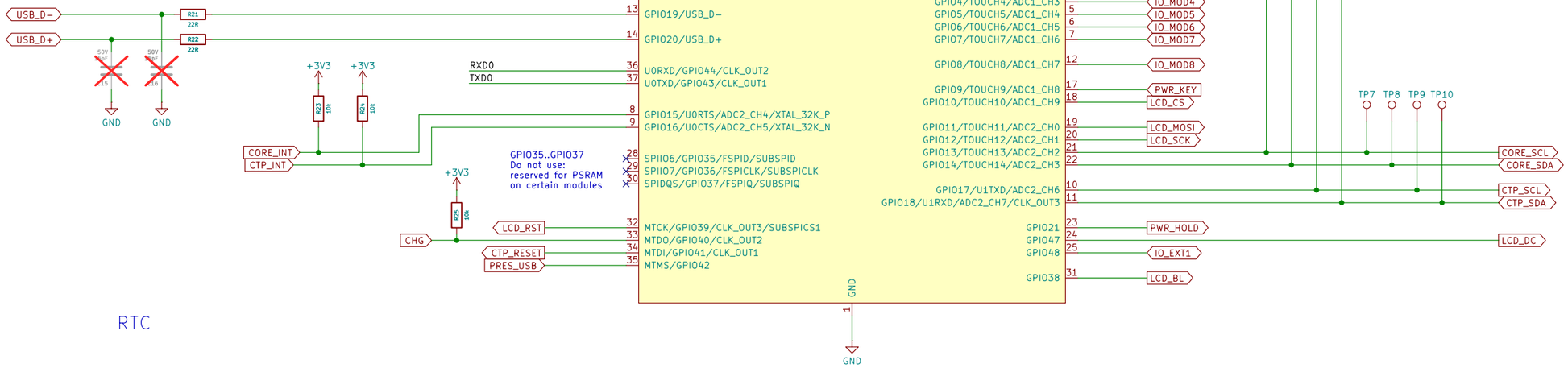
Default configuration at reset
GPIO0(BOOT) -> Pull-up
GPIO45(VDD_SPI) -> Pull-down
GPIO46(BOOT) -> Pull-down

Boot Mode Control
GPIO0 = 1 and GPIO46 = x -> Normal mode
GPIO0 = 0 and GPIO46 = 0 -> Boot mode

VDD_SPI Voltage Control
GPIO45 = 0(default) -> 3.3 V
GPIO45 = 1 -> 1.8 V
Leave un connected

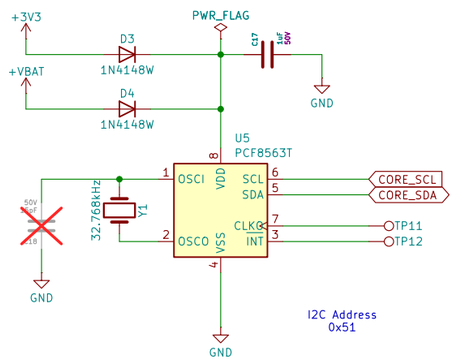
Boot Mode Control
Hold Boot to GND during reset
to enter boot mode

ESP32



GPIO35..GPIO37
Do not use:
reserved for PSRAM
on certain modules

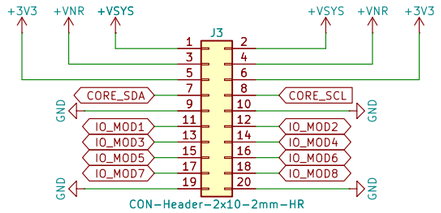
RTC



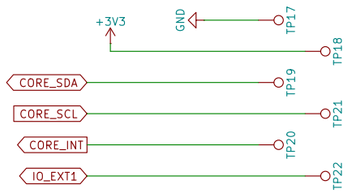
I2C Address
0x51

CORE		
Sheet: /CORE/		
File: core.kicad_sch		
Title: Link Base Board		
Size: A3	Date: 2026-08-28	**Rev: 0.2.0**
KiCad E.D.A. 10.0.5		Id: 3/6

MODULE CONNECTOR



SPARE GPIO + I2C

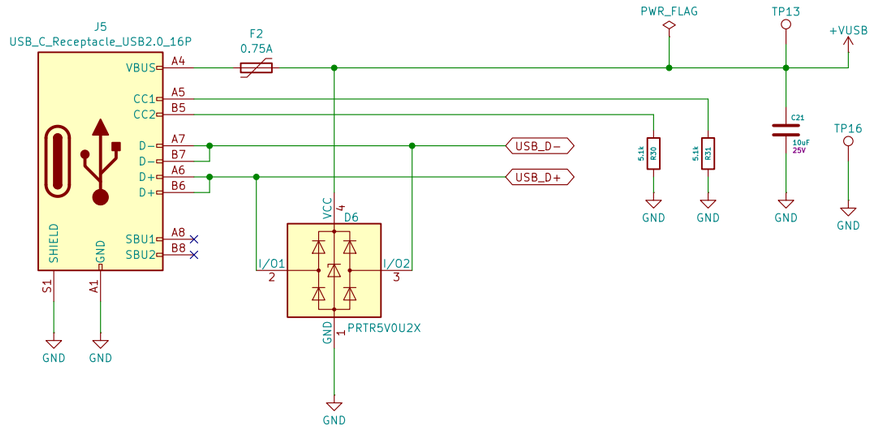


EXTENSION

Sheet: /EXTENSION/
File: extension.kicad_sch

Title: Link Base Board

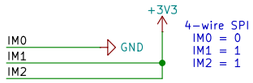
Size: A3	Date: 2026-08-28	**Rev: 0.2.0**
KiCad E.D.A. 10.0.5		Id: 4/6



USB

Sheet: /USB/		
File: usb.kicad_sch		
Title: Link Base Board		
Size: A3	Date: 2026-08-28	**Rev: 0.2.0**
KiCad E.D.A. 10.0.5		Id: 5/6

COMM.CONFIGURATION

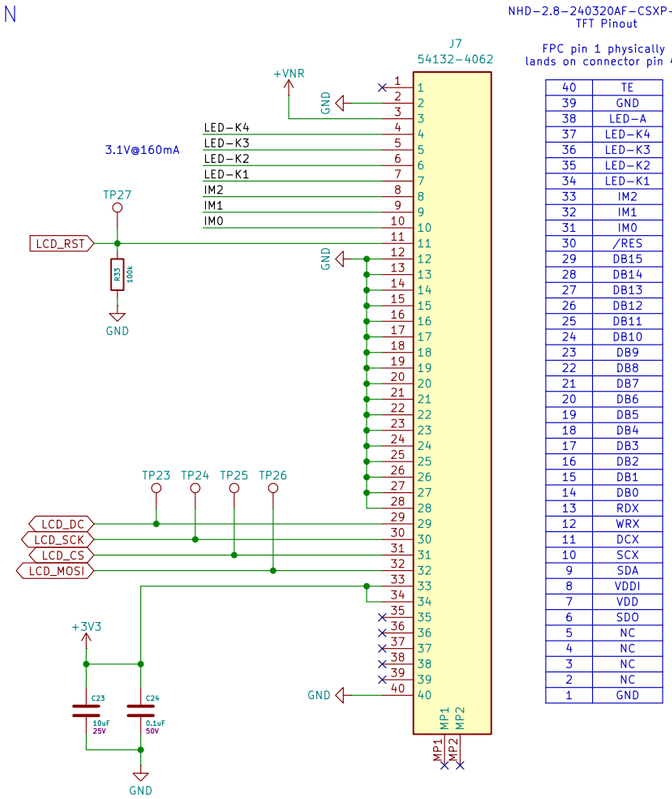


TFT

NHD-2.8-240320AF-CSXP-FCTP
TFT Pinout

FPC pin 1 physically
lands on connector pin 40.

40	TE
39	GND
38	LED-A
37	LED-K4
36	LED-K3
35	LED-K2
34	LED-K1
33	IM2
32	IM1
31	IM0
30	/RES
29	DB15
28	DB14
27	DB13
26	DB12
25	DB11
24	DB10
23	DB9
22	DB8
21	DB7
20	DB6
19	DB5
18	DB4
17	DB3
16	DB2
15	DB1
14	DB0
13	RDX
12	WRX
11	DCX
10	SCX
9	SDA
8	VDDI
7	VDD
6	SDO
5	NC
4	NC
3	NC
2	NC
1	GND

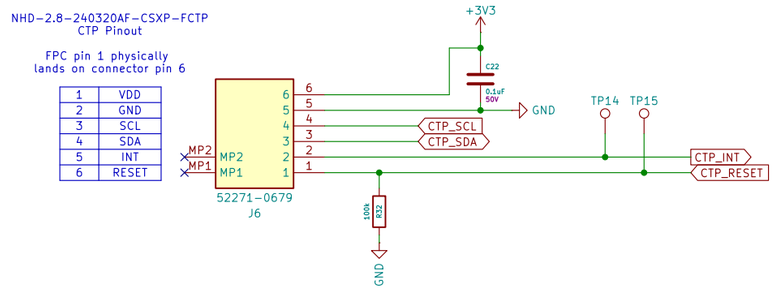


CTP

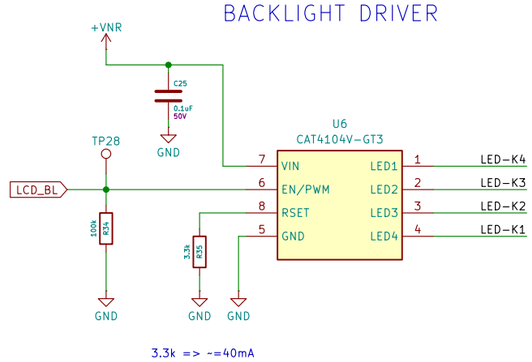
NHD-2.8-240320AF-CSXP-FCTP
CTP Pinout

FPC pin 1 physically
lands on connector pin 6

1	VDD
2	GND
3	SCL
4	SDA
5	INT
6	RESET



BACKLIGHT DRIVER



3.3k => ~=40mA

IHM

Sheet: /IHM/
File: ihm.kicad_sch

Title: Link Base Board

Size: A3	Date: 2026-08-28	**Rev: 0.2.0**
KiCad E.D.A. 10.0.5		Id: 6/6